SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

The issue is **primarily caused by network conditions and application-layer buffering**, rather than the transport protocol itself.

- Video conferencing applications usually use **UDP** because it provides low latency.
- The reported **8% packet loss** and **180 ms jitter** indicate poor network quality.
- The application's buffering mechanism may not be sufficient to compensate for packet loss and jitter, causing voice distortion and frozen video.

b) Justification

Real-time multimedia applications require:

- **Low latency** rather than guaranteed delivery.
- **UDP** is preferred because it does not retransmit lost packets.

- **TCP** would increase delay due to retransmissions and congestion control, making live communication worse.
- Therefore, the transport protocol (UDP) is appropriate, but network conditions and application buffering require improvement.

c) Recommendations

Protocol-Level Improvements

- Continue using **UDP** with RTP/RTCP.
- Enable **Forward Error Correction (FEC)**.
- Implement **Quality of Service (QoS)** to prioritize voice/video packets.
- Improve available bandwidth and reduce congestion.

Application-Level Improvements

- Use **adaptive jitter buffering**.
- Enable **adaptive bitrate streaming**.
- Use better video compression (H.265/AV1).
- Dynamically adjust video resolution based on network conditions.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

DNS Lookup Delay in a Multinational Company

a) Reasons for high DNS resolution time

Possible causes include:

- Centralized DNS server located far from users.
- High DNS server workload.
- Lack of DNS caching.
- Inefficient TTL configuration.
- Multiple recursive lookups.
- Network congestion.

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b) Proposed DNS Architecture

Use the following architecture:

- **Anycast DNS**
  - Multiple DNS servers share the same IP address.
  - Queries automatically reach the nearest server.
- **DNS Pre-fetching**
  - Frequently accessed domain names are resolved in advance.
- **TTL Optimization**
  - Use moderate TTL values (300–600 seconds).
  - Cache DNS responses while allowing timely updates.

This architecture can reduce lookup time from **800 ms to below 50 ms**.

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c) Advantages and Limitations

Advantages	Limitations
Faster DNS lookup	More expensive infrastructure
High availability	More complex management
Better scalability	Requires Anycast routing support
Automatic failover	Cache consistency challenges

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

Distributed DNS for Cloud Service Provider

a) Impact of centralized DNS

Centralized DNS causes:

- High lookup latency.
- Server overload.
- Single point of failure.
- Increased application response time.
- Poor user experience during heavy traffic.

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b) Distributed DNS Design

A suitable design includes:

- Multiple geographically distributed DNS servers.
- **Anycast routing** to direct users to the nearest DNS server.
- Local recursive DNS caches.
- Secondary DNS servers for redundancy.
- Load balancing among DNS servers.

This minimizes lookup delay and improves availability.

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c) Effect of TTL, Anycast, and DNS Caching

Technique	Performance	Fault Tolerance
TTL	Reduces repeated lookups	Faster recovery with lower TTL
Anycast	Routes to nearest server	Automatic failover
DNS Caching	Faster responses	Reduces DNS server load

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction.

Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

SOL:

TCP Latency in Electronic Stock Trading Platform

a) Three TCP-related factors causing latency

1. Flow Control

- TCP limits the transmission rate according to the receiver's window size.
- Small receive windows reduce throughput and increase waiting time.

Effect

- Order acknowledgments are delayed.
- Trading transactions become slower.

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2. Nagle's Algorithm

- Combines small packets before sending them.
- Introduces additional delay.

Effect

- Trade orders wait before transmission.
- Higher acknowledgment latency.

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3. SYN Queue Limitations

- During peak traffic, many new TCP connections arrive.
- The SYN queue becomes full.
- New connection requests are delayed or dropped.

Effect

- Longer connection establishment time.
- Delayed trade confirmations.

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b) Effect on High-Frequency Trading

TCP Factor	Effect on Trading
Flow Control	Slower data transmission
Nagle's Algorithm	Delayed packet delivery
SYN Queue Overflow	Slow connection setup and packet loss

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c) Recommended TCP Configuration Changes

- Disable **Nagle's Algorithm** using **TCP_NODELAY**.
- Increase **TCP receive/send buffer sizes**.
- Increase **SYN queue size**.
- Enable **TCP Fast Open**.
- Use **persistent TCP connections** instead of creating new connections repeatedly.
- Optimize congestion control algorithms (such as **BBR** where appropriate).
- Upgrade network bandwidth and reduce congestion.

Summary

Question	Main Cause	Recommended Solution
1. Video Conferencing	Packet loss, jitter, buffering	UDP, QoS, adaptive buffering, FEC
2. DNS Delay	Centralized DNS, poor caching	Anycast DNS, DNS pre-fetching, TTL optimization
3. Cloud DNS	Single DNS server	Distributed DNS, Anycast, caching, secondary DNS
4. Stock Trading	TCP flow control, Nagle's algorithm, SYN queue	TCP_NODELAY, larger buffers, TCP Fast Open, larger SYN queue

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand